

Scuffle — Arcade Game 9 Design Document (a Stick Fight game)

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Overview

Scuffle is Depth's Stick Fight: a 2D physics brawler for 2–8 players in which stick figures fight on small, dangerous stages with whatever falls from the sky, and the last one standing wins the round. It keeps everything that makes Stick Fight work (one-screen stages, ragdoll bodies, weapons that drop in crates, rounds that last thirty seconds, levels that try to kill you) and adds the things a group that plays it for a year wants: more players, more worlds, a level editor with sharing, trinkets that make each stick a little different, mutators, mid-round events, objective modes, co-op boss arenas, and a gauntlet.

What it keeps from Stick Fight

- Stick figures with full ragdoll physics: every death is a tumble, every hit is a flail, every bullet moves a body.
- One-screen stages that are the real opponent: lava, saws, spikes, snakes, collapsing floors, black holes.
- Weapon crates from the sky at a timer; everyone starts with fists.
- Fists that punch, block bullets with a timed punch, and knock people into hazards.
- Rounds of about thirty seconds; first to a score wins the match; the stage changes every round.
- Chaos as the design: the balance is that nothing is balanced for long.

What Scuffle adds

1. **Eight players** on bigger stages, with camera zoom that follows the pack.
2. **Six worlds** from Depth (the Nautilus, the Cave, the Reef, Atlantis, the Void, the Salon) with their own hazards, and a stage generator alongside hand-built stages.
3. **Trinkets**: one small perk each stick picks per match (a double jump, a thick skull, a lucky crate).
4. **Mutators**: lobby rules that change the physics or the arsenal for a match (low gravity, bullets bounce, everything's a snake).
5. **Events**: mid-round surprises (the floor floods, the kraken reaches up, the lights go out).

6. **Gear beyond guns:** the grappling hook, the shield, the jetpack, the decoy.
7. **Modes:** teams, king of the plank, the egg, hot potato, co-op boss arenas, the gauntlet.
8. **A level editor** with instant play and sharing by code.

Design pillars

1. **Thirty seconds of panic, then laughter.** Rounds are short; the stage is the joke; death is a punchline, never a penalty.
2. **Physics over stats.** Everything is a body with mass. A punch moves you; a shotgun moves you more; a black hole moves everyone.
3. **Easy to join, impossible to finish learning.** Two buttons and a stick play it; the ninetieth round still surprises.
4. **Depth's world, Depth's ink.** Stick figures drawn in the ink style, stages from the game's locations, weapons that are Depth's relics with a trigger.

Session. A match is first to 10 rounds (lobby: 5 / 10 / 20), 10–20 minutes; 2–8 players, bots fill. Score pays arcade tokens; tokens buy skins and hats or spin the crate.

Tone. Fast, loud, inked. The stick figures have one face (two dots and a line) and it conveys everything.

The feel

Everything in Scuffle is a physics body, including you. The stick figure is an active ragdoll: a torso, a head, two arms, two legs, driven by motors toward a target pose while you're alive and limp the moment you aren't. The movement code's job is to make a body that's always slightly out of control feel exactly controllable.

Movement

Action	Input	What happens
Run	Stick or A/D	8 m/s with a lean; running into a wall is a small bonk
Jump	A or space	A 3 m hop; hold for height; the body tucks
Wall climb	Hold toward a wall	Hands grab; climb at 3 m/s for 1.5 s, then slide
Wall jump	Jump from a wall	A push off at 45 degrees; chainable

Action	Input	What happens
Duck	Down	Head under the shot line; can't move
Dive	Down in the air	A headfirst dash; lands prone; a prone stick is harder to hit and slower to get up
Get up	Any input while ragdolled	0.4 s if the body is still; longer if it's tumbling

There is no double jump by default (a trinket gives it). Knockback is real: a shotgun blast is 6 m of flight; a punch is 1.5 m; falling 10 m stuns for 0.5 s.

Fists

- **Punch** (fire with no weapon): a jab with the lead arm, 10 damage, 1.5 m knockback, 0.25 s. Three in a row is a haymaker (20, 3 m).
- **Block**: a punch thrown at a bullet in the 0.15 s before it lands deflects it (it flies off at the punch's angle). Blocking a rocket deflects the rocket. Blocking is the skill ceiling and the reason unarmed is never hopeless.
- **Grab** (hold fire while touching): hold a stick or a crate; throw it with a direction. A thrown stick is a projectile (15 damage to both). Grabbing a weapon's owner can pull the weapon loose.
- **Kick** (fire while diving): 15, 4 m knockback, the thing you do to someone standing on a ledge.

Health

100 HP; most bullets do 25–40; a melee weapon 30–50; hazards kill outright. No regeneration during a round. Low health shows as a wobble and a red tint on the stick; dead sticks are ragdolls that stay on the stage as obstacles and can be thrown.

Weapons in hand

- You hold one weapon; picking up a second drops the first. Weapons have ammo and are dropped empty (they're still a thrown object: 10 damage).
- Aim is a free 360-degree arm; recoil moves the body (a minigun pushes you backward; a sniper rocks you).
- Weapons are bodies too: a dropped shotgun falls, slides, and can be shot off a ledge.

Why it's funny

The ragdoll makes everything expressive: a stick that's shot spins; one that's punched flails; one that's frozen tips over whole; one on fire runs in circles (an AI that takes over the body); one that falls into lava reaches back up for a half second. Every weapon has a wrong way to use it that's better than the right way, and the stage is always about to collapse.

Rounds and matches

A round

1. A stage loads (from the match's world rotation or the editor's playlist); sticks spawn at the stage's spawn points, evenly spaced, facing inward.
2. A 1 s countdown (the sticks stretch).
3. Fight. The first weapon crate falls at 3 s, then one every 5 s, from the sky, on parachutes, at random points over the stage (the stage can mark "crate zones"). A crate opens on touch; it's a weapon, or sometimes gear, or sometimes a snake.
4. The round ends when one stick is standing (or one team). Standing means alive: a ragdolled living stick counts.
5. **The wall.** At 45 s, if more than one stick lives, the stage starts to kill: by world, lava rises, the walls close, the floor crumbles from the edges, the lights go out and the hazards don't. By 60 s only the center is safe; at 70 s nothing is.
6. Winner's pose (1.5 s), score, next stage.

The match

- First to 10 round wins (5 / 10 / 20). Draws (everyone dies at once) give nobody a point and are a story.
- The stage rotation draws from the chosen worlds with no stage twice in a match; the last round of a match is always a **finale stage** (the biggest in the world, with the wall at 30 s).
- Players can join and leave between rounds; bots fill to the lobby's count and leave when a human joins.

Scoring and tokens

Score	Points
Round won	100
Kill	20; +10 by hazard knock (you punched them into it); +20 by block-deflect
Survived to the wall	10
Match won	300
Stories (a triple kill with one rocket, a win with fists only, a win while on fire, surviving a black hole)	50 each

Tokens: 10 per match, 1 per 50 score, 20 per match won, 25 per first story. The leaderboard shows round wins; the end-of-match screen shows the kill feed's best moments replayed (a 10 s ragdoll replay of the round's most physics).

Pace

A 10-round match with 6 players runs 12–15 minutes with the wall; an 8-player match has more draws and more laughter. The design target is a round every 40 s on average.

Weapons

Forty-eight weapons in crates, all physics objects, all with a wrong use. They're Depth's arsenal with triggers: gas pistols and harpoons from Red Tide, the Nautilus's brass, the Flats dealer's bottles, and a few things nobody asked for. Weights drop by world so every match's arsenal leans a different way.

Guns

Weapon	Damage	Rate	Ammo	Knockback	The wrong use
Gas pistol	25	3/s	12	Small	Shoot the ground under someone's feet (it breaks)
Twin pistols	25 × 2	5/s	24	Small	Aim in two directions (left stick and right stick)

Weapon	Damage	Rate	Ammo	Knockback	The wrong use
Needler	12	12/s	60	Small, constant	Hold it down on a wall to climb by recoil
Carbine	40	2/s	10	Medium	A headshot (the head is a separate body) is 80
Scatter gun	15 × 8	1/s	6	Big	Point it down and fly
Harpoon rifle	70	0.7/s	5	Big; pins to walls	Pin a stick to the ceiling
Sniper	100	0.5/s	3	Rocks you	Shoot through three sticks in a line
Minigun	10	25/s	200	Pushes you backward constantly	A jetpack if you aim down
Grenade launcher	60 area	1/s	6	Big	Bounce it off two walls
Rocket launcher	90 area	0.5/s	3	Huge	Rocket jump; block-deflect it back
Bazooka of Bees	5 × many	1/s	4	None	Bees follow the nearest stick, which is sometimes you
Flamethrower	8/s while in the cone	hold	100 units	None	Set the stage on fire (wood burns)
Ice gun	0	hold	100	None	Freeze a stick (it's a statue for 2 s; it shatters into a ragdoll); freeze lava into a bridge
Tesla gun	30 chain	1/s	8	Small	Chains through water (and anyone standing in it)
Snake gun	20 (a snake)	2/s	10	None	Snakes bite whoever they land on, then slither; a stage can fill with snakes
Black hole gun	Pulls	1	1	Everything	Everything within 6 m is pulled in, including crates, weapons, and you; the hole lasts 4 s
Bubble gun	0	2/s	20	Lifts	Bubbles carry anything they hit upward for 3 s; pop them with a punch
Gravity gun	0	hold	100	Flips	Reverses gravity for whatever it's pointed at
Laser	50/s beam	hold	3 s	None	Reflects off ice and mirrors; cuts ropes

Weapon	Damage	Rate	Ammo	Knockback	The wrong use
Chum cannon	0	1/s	5	Small	Fires chum; the stage's fish (Reef, Void) come to it
Net gun	0	1/s	4	None	Holds a stick 3 s; nets a crate before it lands
Boomerang pistol	30	1/s	infinite	Small	It comes back; it hits you if you miss
Confetti cannon	5	1/s	10	Big	Pure knockback; a party
Portal gun	0	2 shots	2	None	Two portals; sticks, bullets, and lava go through

Melee

Weapon	Damage	Swing	The wrong use
Boarding axe	50	Slow	Throw it (it spins; 60)
Cutlass	35	Fast	Deflects bullets on a timed swing
Trident	40, reach	Medium	Pole vault
Frying pan	30	Medium	Blocks everything; cooks bees
Pool cue	25	Fast	Breaks into two halves (two weapons)
Baseball bat	40	Medium	Bat a grenade back
Sledgehammer	80	Very slow	Breaks any floor tile
Whip	20, long reach	Fast	Grab a ledge or a stick and pull
Fish	15	Fast	A slap; it's a fish
Oar	30	Medium	Row through water twice as fast

Thrown and placed

Item	Effect	The wrong use
Limpet charges (3)	Stick to anything; 60 area on a 2 s fuse	Stick one to a stick

Item	Effect	The wrong use
Sticky bombs (3)	Same, 1 s fuse, bounce once	Stick one to a crate in the air
Ink bomb (2)	Blinds the stage for 3 s (everyone's screen)	Throw it at the start of the wall
Flashbang (2)	White screen 1.5 s	Blind yourself and block anyway
Bear trap (2)	Holds a stick that steps on it until punched free	Put it under a crate zone
Turret	A placed gun that shoots the nearest stick for 10 s	Place it on a moving platform
Mine (3)	70 when stepped on	Throw it under someone jumping
Banana (3)	Slips; 0 damage; a 0.8 s ragdoll	Peel order matters not at all
Beehive	Bees for 8 s	Roll it
Jar of snakes	Four snakes	Open it on yourself by mistake

Crates

- Crates fall every 5 s at random crate zones, on parachutes (shoot the parachute and the crate drops faster; it crushes a stick it lands on for 30).
- Weights by world (the Reef drops more chum cannons and snakes; the Void drops more black holes and gravity guns; the Salon drops more frying pans and fish). A match's **arsenal** can also be chosen in the lobby: Classic (guns), Melee only, Chaos (the strange ones), Snakes (it's all snakes), Random.
- One crate in ten is **gear** (next section) and one in thirty is a **trap** (a snake, a bee hive, a flashbang that goes off on open).
- A crate can be punched across the stage, shot (it opens where it lands), or grabbed and thrown as a weapon (20).

Weapon balance

The rule is that every weapon wins a one-on-one in some situation and loses in another: the sniper owns long sightlines and loses in a corridor; the scatter gun is the reverse; the minigun wins if it starts firing first and is useless if punched; the black hole wins everything once and then there's a black hole. Weights, damage, and ammo are in `scuffle_weapons.json` and are the first thing a balance pass touches.

Levels

A stage is one screen (two for eight players, with the camera zooming to fit the living sticks), built from tiles that are physics bodies: platforms, ropes, moving pieces, breakable pieces, and hazards. Six worlds, 40 hand-built stages each, a generator for endless ones, and the editor for everyone else's.

The six worlds

World	Look	Hazards	Signature stages	The wall
The Nautilus	Brass, rivets, pipes, portholes, the engine room	Steam vents (knockback), pistons (crush), the propeller (shreds), electrified rails, flooding bulkheads	The Engine Room (pistons on a rhythm), The Torpedo Tubes (tubes fire sticks across the stage), The Salon Window (the glass breaks at 20 s)	Bulkheads flood from the bottom
The Cave	Stalactites, glow-mould, pools, the Lobster's crystal	Falling stalactites (triggered by gunfire), acid drips, crystal (shatters into shrapnel), slipstreams (one-way currents that carry sticks), the toad's tongue from a pool	The Chimney (vertical; sticks climb), The Slipstream (everyone's moving), The Lobster (a giant claw sweeps on a timer)	The ceiling comes down

World	Look	Hazards	Signature stages	The wall
The Reef	Coral, sand, sunlight, the surge	Reacher coral (grabs), eel holes (bite), urchins, the surge channel (a current every 10 s), sharks in the water below	The Bommie (a pinnacle), The Flats (open; no cover; sharks), The Drop-off (a cliff into blue water)	The tide comes in
Atlantis	Marble, columns, the god-pool, the plaza	Grates (the Wyrms strike from them), tuna crossing (a ram), the sluice (floods a terrace), falling columns	The Plaza (the tuna lane), The Baths (slippery; a hypocaust boils), The Cisterns (grates everywhere)	The city sinks
The Void	Black rock, vents, the abyss on one side	The abyss (fall and you're gone), brine pools, the leviathan's lure (a light that pulls), low gravity pockets, the sand worm's line	The Rim (the worm takes the edges), The Overlook (a ledge over nothing), The Reactor (vents on a timer)	The abyss widens
The Salon	The Sodden Gull: the bar, the pool table, the stage, the yard	Thrown bottles from the crowd, the dartboard (darts fly), the pool balls, the bouncer (throws anyone who touches the bar out), the dog	The Bar (the counter is the stage), The Pool Table (the balls are hazards), The Yard (mini golf windmills)	Closing time: the bouncer clears the room

Stage pieces

- **Platforms:** stone (permanent), wood (burns, breaks under a sledgehammer), ice (slippery; melts under fire), glass (breaks under weight), rope bridges (cut with a laser or a cutlass), chains (swing).

- **Movers:** elevators, conveyors, rotating beams, pendulums, pistons, the slipstream, the tide.
- **Breakers:** crumbling floors (a timer after the first step), explosive barrels, the salon's chandelier.
- **Water:** sticks swim slowly, can't shoot guns underwater (harpoons and tesla work), and drown after 8 s; electricity in water hits everyone in it.
- **Wrap:** some stages wrap left-right (a stick off one edge comes in the other); the Void's never do.

The generator

For endless play: it picks a world, lays 3–7 platforms on a grid with the world's pieces, guarantees every spawn can reach every other, adds 1–3 hazards and 1–2 movers, and runs a reachability check with the real movement code (the parkour side's `--verify` approach). Generated stages are marked and can be saved to the editor if the group liked one.

Finale stages

Each world has three finales: twice the size, the wall at 30 s, and a set piece (the propeller starts, the Lobster rises, the kraken reaches up from the drop-off, the Wyrms surfaces in the plaza, the leviathan's lure appears, closing time with the whole bar thrown in).

New features

Trinkets

Each stick picks one trinket in the lobby for the match (or lets the game pick). A trinket is one small, visible edge; everyone can see what everyone took, so the counter-play is open.

Trinket	Effect	The cost
Spring Heels	Double jump	Shorter first jump
Thick Skull	Headshots do normal damage	Slower get-up
Lucky Crate	The first crate of each round lands on you	It also lands <i>on</i> you (30) if you don't move
Long Arms	Punch reach +50%	Punches are 0.1 s slower

Trinket	Effect	The cost
Magnet Palms	Wall climb never slides	Can't dive
Cat Legs	No fall stun	Lighter: knockback +20%
Big Lungs	Drown time doubled; shoot underwater	Slower swim
Quick Draw	Pick-ups are instant and the first shot has no spread	Ammo -20%
Pack Rat	Carry two weapons (swap)	Run speed -10%
Thick Coat	Fire and ice don't stick	Can't climb
Loud Mouth	Your taunts (emotes) push sticks within 1 m	Everyone can hear where you are in Blackout
Second Wind	Once a round, survive a killing hit at 1 HP	You're on fire for 2 s after
Snake Charmer	Snakes don't bite you	Everyone else's snakes come to you
Deadweight	Knockback -50%	Can't be bubbled or lifted; you fall straight through bubbles
None	A clean stick	Round wins pay +10

Mutators

Lobby rules that change the match. Any number can stack; a Random mutator picks one per round.

Mutator	What changes
Low Gravity	Jumps are twice as high; everything floats
Moon Shot	Bullets are slow and visible; blocking is easy and so is dodging
Ricochet	Every bullet bounces once
Big Heads	Headshots are easy; heads are physics balloons

Mutator	What changes
Infinite Ammo	Nothing runs dry
One Hit	Everything kills; fists included
Ragdoll Royale	Everyone is always ragdolled; movement is flailing (the game's hardest mode and its funniest)
Snakes	Every crate is snakes
Hot Potato	A bomb that passes by touch; it goes off at 10 s
Blackout	Lights off; muzzle flashes and hazards glow
Giants	Every stick is twice the size (and the stage isn't)
Tiny	Every stick is half the size; hazards are the same
Mirror	The stage is mirrored each round; wrap is on everywhere
Vampire	Kills heal 50
Sudden Wall	The wall starts at 15 s
Pacifist	Guns don't drop; melee and gear only; hazards do the killing
Fast Forward	Everything at 1.3× speed

Mid-round events

One in four rounds has an event at a random second between 10 and 30, announced by a 1 s tell:

Event	What happens
The Flood	Water rises to half the stage for 10 s; guns stop; tesla is king
The Reach	The kraken's arm comes over the edge and sweeps the lowest platform
Lights Out	5 s of dark

Event	What happens
Crate Rain	Twenty crates at once
Earthquake	Every breakable breaks; every mover stops
Swap	Every stick swaps position with another
The Dog	The alley dog runs through and bites whoever's closest
Gravity Flip	Up is down for 4 s
The Bouncer	The Salon's bouncer throws the stick with the most round wins into the hazards
Fish Storm	Fish fall from the sky; they're weapons; they're everywhere

Gear

One crate in ten is gear: an item used from a second slot (the gear button) that lasts the round.

Gear	What it does
Grappling hook	Fire a line at anything (10 m); reel in; swing. The best movement in the game and the best way into a hazard
Shield	A buckler on the off arm; blocks bullets from the front while held; melts to fire
Jetpack	3 s of thrust, recharges on the ground; burns anyone under you
Decoy	Drops a stick-shaped dummy that turrets, bees, and snakes prefer
Parachute	Slow falls; steer; the way to survive the abyss once
Spring	A placed trampoline
Rope	A placed rope between two points: a bridge or a tripwire

Gear	What it does
Mirror	A placed plate that reflects lasers and bullets
Balloon	Tied to you: you float up until it's popped
Fishbowl helmet	Shoot underwater; breathe; +10 head HP

Taunts

Six emotes (point, laugh, flex, bow, cry, the fish slap) on the taunt wheel. A taunt is a 1 s animation during which the stick's guard is down. Loud Mouth makes them a weapon.

Modes

Mode	Players	Rule	Why
Scuffle (classic)	2–8	Last stick standing; first to 10	The game
Teams	4–8	2v2, 3v3, 4v4, or 2v2v2v2; friendly fire on (it's funnier)	Shouting at a friend
King of the Plank	3–8	A marked platform scores 1 per second to whoever stands on it alone; first to 60; the plank moves every 20 s	Fights over one spot
The Egg	3–8	Hold the egg (a physics object that breaks if dropped from height) for 30 s total; the holder can't shoot, only punch	Keep-away with a ragdoll
Hot Potato	3–8	A bomb passes by touch; it detonates at 10 s; last stick wins	Tag with consequences
Hunt	4–8	One stick is the Shark (double HP, a harpoon, no crates); the rest have fists and crates; the Shark wins by eating everyone before the wall; a kill makes you a Shark too	Infection

Mode	Players	Rule	Why
Duel	2	Best of 7 on finale stages; both pick a weapon before the round from the same three	A real fight
Boss Arena (co-op)	2-8	Everyone against a Depth boss on its stage: the Lobster (claws sweep; crystal falls), the Kraken (arms over the edge; the eye is the weak point), the Wurm (surfaces from grates), the Sun God (fire beams), the Goliath (inhales), the Bouncer (throws); bosses have phases and the stage changes with them; players respawn on a timer until the wall	A raid
The Gauntlet (co-op)	1-8	A run of 20 generated stages with no other sticks: traps, movers, hazards, and a timer; a group races the clock and each other's deaths; stages get worse; a leaderboard per world	Parkour with guns
Chaos	2-8	Random mutator every round, Random arsenal, events every round	The party setting
Custom	2-8	Any mutators, any arsenal, any playlist (hand-built, generated, editor), any score to win, the wall on or off	Whatever the group wants

Bots

Bots play every mode with three levels (Stumble, Scrap, Sharp) and personalities (a bot that always rushes, one that camps crates, one that only uses melee, one that taunts). They fill lobbies, they're named like pub regulars, and in Boss Arena they're competent enough to carry a duo.

Join-in-progress

Anyone can join a match between rounds; a spectator can watch any round with the replay camera.

The level editor

Stick Fight's long life came from its Workshop. Scuffle's editor is built in, instant, and shares by code so that no Steam is needed.

Building

- A grid of tiles the size of a stick's torso; the stage is 32×18 tiles (or 64×18 for eight players), with the camera showing it all.
- A palette by world: platforms (stone, wood, ice, glass, rope, chain), movers (elevators with waypoints, conveyors, rotators, pendulums, pistons on a timer), hazards (every world's), water, breakables, crate zones, spawn points (2–8), the wall's style, wrap on or off, and decoration.
- Properties per piece: timers, speeds, ranges, trigger links (a switch opens a door; a crate landing starts a piston).
- **Play now:** one key drops you into the stage with bots; the editor stays open; edits apply live.
- **The check:** the stage won't save until every spawn can reach every other spawn in the reachability test (the real movement code), so no stage is unwinnable by accident.

Sharing

- A stage exports as a short text code (the tile data compressed and encoded) that fits in a chat message; paste it into the lobby and it's in the playlist. No server, no account.
- A **pack** is up to 20 stages in one code.
- The arcade profile keeps a library of stages and packs with names and authors; a stage played in a match is offered to everyone in the lobby to save.
- Stages carry a world tag so the wall and the crate weights know what to do, and a difficulty tag the generator's checker estimates (hazard count, platform area, movers).

Scripting, lightly

No code. Triggers and links cover what people build: switches, timed doors, a trap that fires when a crate lands, a platform that falls after the third step, a boss-arena piece that spawns a hazard on a timer. An **event tile** lets a stage call one of the mid-round events at a chosen second.

Built-in packs

The six worlds' 40 stages each are built in the same editor and ship as packs, so the first thing a player can do is open one and see how it works.

Skins, hats, and the crate

A stick is a color and a hat. Colors are free (eight, by player slot, and the lobby lets you pick). Skins change the stick's line style and body; hats sit on the head (a separate physics body, so hats fly off on a headshot and can be picked up by someone else). Tokens buy them; the crate hatches them; nothing changes a stat.

The token shop (10 skins, 10 hats)

Skin	Cost	Look
Ink Blot	50	A thicker, wobblier line
Chalk	60	White chalk on the stage's black
Brass	80	Riveted limbs
Skeleton	100	Bones for lines
Ghost	100	Translucent
Wire	120	A bent-coat-hanger stick
Neon	120	Glow in Blackout (a disadvantage, worn on purpose)
Noodle	150	Limbs are wet noodles (purely visual; the ragdoll is the same)
Dotted	150	A dotted-line stick
Golden	200	Gold leaf

Hats: the Sailor Cap (50), the Diver's Helmet (80), the Top Hat (80), the Crown (100), the Traffic Cone (100), the Frying Pan (120), the Jellyfish (150), the Tiny Kraken (150), the Lighthouse (200), the Fish (200: it's a fish).

The crate (40 items)

One token, one crate (a weapon crate on a parachute that opens on the menu). Duplicates give a banana. Odds: Common 60% (15, 4% each), Rare 25% (10, 2.5%), Super Rare 12% (10, 1.2%), Special 3% (5, 0.6%).

Tier	Items
Common (15)	Hats: Beanie, Bandana, Bucket, Paper Hat, Headband, Sombrero, Beret, Hard Hat, Party Hat, Flower, Antenna, Goggles, Monocle, Eyepatch, Bow

Tier	Items
Rare (10)	Skins: Crayon, Marker, Pencil Sketch, Blueprint, Rust, Barnacled, Candy Stripe, Pixel (the parkour's), Stitched, Wet Paint (drips while running)
Super Rare (10)	Hats: the Matriarch (an orca), the Sand Worm Puppet (yes, again), the Goat, the Lantern (lights Blackout), the Bathtub, the Slot Machine, the Dealer's Visor, the Nautilus, the Reacher Coral (grabs nothing), the Chandelier
Special (5)	Skins: Red Tide (bleeds ink when hit), Lightning (the Tesla line), Molten (sets wood on fire when standing on it; visual only, it does not), the Blobfish (a blob with limbs), the Golden Kidney (one gold kidney, sewn on)

Skins and hats are data (`scuffle_cosmetics.json`); the crate shares the arcade token.

Presentation, physics, technology, and netcode

Look

- Depth's ink in 2D: stick figures are three-pixel black lines with round joints on parchment-toned stages; the worlds are drawn in the expedition backgrounds' style (inked, hatched, a little painterly) with the stage's physics pieces outlined heavier so what's solid is obvious.
- Deaths are ink: a stick that dies goes limp and grey; blood is an ink splash; fire is ink that curls; ice is a white hatch.
- Weapons are drawn as Depth's relics: the harpoon rifle from Red Tide, the frying pan from the Gull, the Tesla gaff.
- The camera frames all living sticks with a minimum and maximum zoom; replays use a free camera.

Physics

- A 2D rigid-body engine (Box2D-style, written in the codebase or vendored) at 120 Hz with fixed steps; sticks are 6-body ragdolls with motor joints driven toward animation poses (run, jump, climb, aim, taunt) by strength that drops to zero on death, stun, or freeze.
- Everything on the stage is a body: crates, weapons, hats, snakes, bees, bubbles, water (a fluid approximation: buoyancy volumes), ropes (chains of segments), breakables (bodies with health).
- Determinism matters here more than in any other arcade game, because physics is the game: fixed-point or carefully ordered float math so that the host and clients agree.

Netcode

- Host-authoritative with **rollback for inputs** (the fighting-game approach): clients predict their own stick from inputs, the host confirms, mismatches roll back and replay up to 6 frames. Eight sticks and a few hundred bodies is small enough for this to feel local at 100 ms.
- Physics on the host is the truth; clients run the same deterministic simulation from the same inputs and only correct on divergence.
- The shared arcade layer handles lobby, join codes, LAN, relay, and profiles. Bots run on the host.

Performance

A stage has under 500 bodies at the worst (Crate Rain plus snakes plus bees); the target is 120 Hz physics and 60 fps on a laptop with eight players.

Replays

Every round's inputs are recorded; the end-of-round replay re-simulates the last 10 s with a slow-motion camera on the kill; a match's best three rounds are saved to the profile as replay files that re-simulate on demand (small, because they're inputs).

Sound design

Scuffle's sound is percussion: every impact is a hit, every stick has a yelp, and the stage has a rhythm you can play to.

- **Sticks:** a yelp per hit (pitched per player color), a grunt per punch, a scream per fall, a squeak when frozen, a sizzle when burning, a gurgle underwater, a pop when a head comes off. Taunts have a voice each.
- **Weapons:** every weapon has a distinct fire, hit, pickup, and empty sound; the minigun's spin-up, the black hole's pull (the mix bends into it), the bubble gun's giggle, the snake gun's hiss, the confetti cannon's party.
- **Stages:** the engine room's pistons on a beat, the cave's drips and the slipstream's rush, the reef's surge channel on a count, the plaza's tuna whoosh, the Void's silence and the worm's rumble, the Salon's crowd and jukebox. Hazards have tells you can hear a half second before they fire.
- **The wall:** each world's wall has a rising sound (water, grinding rock, the tide, the city's groan, the abyss's tone, the bouncer's whistle).
- **Events:** a 1 s tell each (a splash, a tentacle's slap, a fuse, a thunderclap, a bark).
- **Rounds:** a countdown tick, a round-win sting, the replay's slow-motion low-pass, the match-win fanfare.
- **Music:** a fast brass-and-drum loop per world that drops out during the wall and comes back as a riff on the win; Boss Arena has the boss's theme from the expeditions; the Gauntlet has a ticking clock under its loop.

Build order, data, tests, and open questions

Data files

`scuffle_weapons.json` (the 48 weapons, crate weights by world and arsenal),
`scuffle_gear.json`, `scuffle_trinkets.json`, `scuffle_mutators.json`,
`scuffle_events.json`, `scuffle_worlds.json` (pieces, hazards, walls, finale set pieces),
`scuffle_stages/` (the 240 built-in stages as editor codes), `scuffle_bosses.json`,
`scuffle_bots.json`, `scuffle_modes.json`, `scuffle_cosmetics.json`.

Build order

Stage	Work	Gate
1	The ragdoll: a stick that runs, jumps, climbs, punches, blocks, and dies on one stone stage; deterministic physics	Two sticks fight with fists and it's funny

Stage	Work	Gate
2	Crates, the first 12 weapons, the wall, rounds and matches, bots	Four bots finish a 10-round match
3	The editor with the reachability check and codes; the Nautilus world's 40 stages built in it	A stage round-trips through a code
4	Multiplayer: rollback netcode through the shared layer; eight players on LAN	Eight players, 100 ms, no visible correction
5	The other five worlds and their stages; the generator; finale stages	Every world's wall and set piece works
6	The full arsenal, gear, trinkets, mutators, events	<code>--scuffle-arsenal</code> shows every weapon wins somewhere
7	Modes: teams, objectives, Hunt, Duel; the Gauntlet	Each mode finishes with bots
8	Boss Arena: six bosses with phases	A duo beats the Lobster
9	Cosmetics, crate, replays, sound, internet play, balance pass	All targets met or logged

Tests

- `--scuffle-sim <players> <rounds> <arsenal>` runs bots and reports: round length, kills by weapon, deaths by hazard, draws, and the wall's share of deaths.
- `--scuffle-arsenal` fights every weapon against every other on three stage shapes (open, corridor, vertical) and asserts every weapon has a shape it wins on.
- `--scuffle-verify <stage code>` runs the reachability check; `--scuffle-verify-all` on the 240 built-ins.
- `--scuffle-determinism <seed>` runs the same inputs twice and asserts identical physics.

Balance targets

- Round length 25–45 s with four players; 30–60 with eight; draws under 8%.
- Deaths: weapons 55%, hazards 30%, fists and throws 10%, the wall 5%.
- No weapon over 12% of kills across a 100-round sample on Random arsenal.

- Every trinket picked between 3% and 15% of the time by Sharp bots (none dominant, none dead).
- A Sharp bot beats a new human about 60% and loses to a good one about 70%.

Open questions for the designer

- ☐ Friendly fire in Teams: on as written, or a lobby toggle?
- ☐ Should the wall be on by default in Custom, or off (Stick Fight's rounds can run long without one)?
- ☐ Is rollback netcode worth the build cost over plain host-authoritative with prediction? It's the difference between the game feeling local and feeling fine.
- ☐ Should editor codes be shareable across the arcade (a Scuffle stage inside The Trawl's lobby chat, say), or kept to Scuffle's lobby?